



Twinkl

An inner compass that helps users align behavior with long-term values

Capstone 5 Project Proposal · April 2026

- Thesis: Twinkl is building a value-alignment engine for journaling - A system that compares stated values with lived behavior over time.

Problem, Users, and Market Gap

The Core Tension

Ambitious people often know what matters to them - yet their weeks quietly drift away from those priorities. The problem is not a lack of reflection. It is a lack of **accountability to declared values**.

Existing AI journals help users log feelings, summarize themes, and maintain habits. They rarely maintain an explicit self-model of the user's values, priorities, and trade-offs.

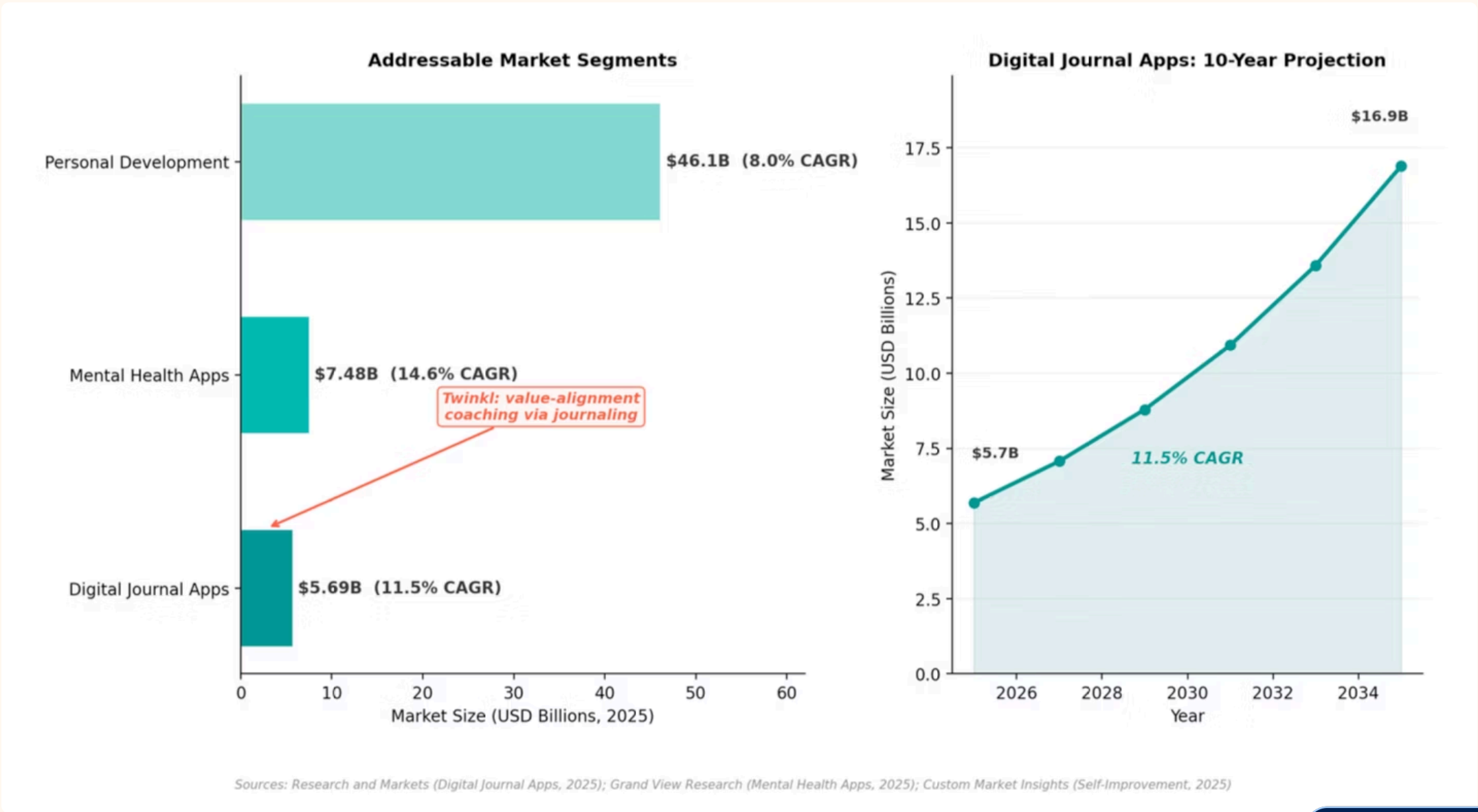
Target Users

Twinkl starts with knowledge workers in transition, people already trying to balance work, health, relationships, and identity under pressure.

These users already pay for journaling, coaching, and reflective productivity tools, which makes them a credible early market.

MARKET OPPORTUNITY

As digital journaling grows, the next frontier is not better logging, but better guidance: tools that turn reflection into ongoing alignment with personal values and priorities.



Why Twinkl Is Not Another AI Journal

Most AI journals summarize entries after the fact. Twinkl starts with a declared values profile and checks whether behavior stays aligned with it over time.

Dimension	Typical AI Journal	Twinkl
Starting Point	Starts from entries and infers themes afterward	Starts with a declared profile of values and goals
Core Job	Summarizes moods, themes, and habits	Evaluates alignment between stated priorities and lived behavior
Memory	Uses short-term context or retrospective summaries	Maintains a persistent self-model with value weights, tensions, and evidence
Feedback Style	Generic prompts and wellness reflections	Evidence-grounded weekly reflections on trade-offs, drift, and alignment
Restraint	Nil, Responds conversationally by default	Uses uncertainty gating and can hold back when confidence is low

Research-Grounded Alignment Coaching



Schwartz's Theory of Basic Human Values

A 10-dimensional model of competing priorities, so alignment is measured across multiple values rather than flattened into a single score.



Best-Worst Scaling

Forced-choice method for eliciting values during onboarding. It captures trade-offs more clearly and reduces the “everything matters” problem common in self-reporting.



Synthetic Longitudinal Data

Temporally grounded synthetic personas because no public dataset pairs journal trajectories with per-value alignment labels.



LLM Judge with Human Benchmarking

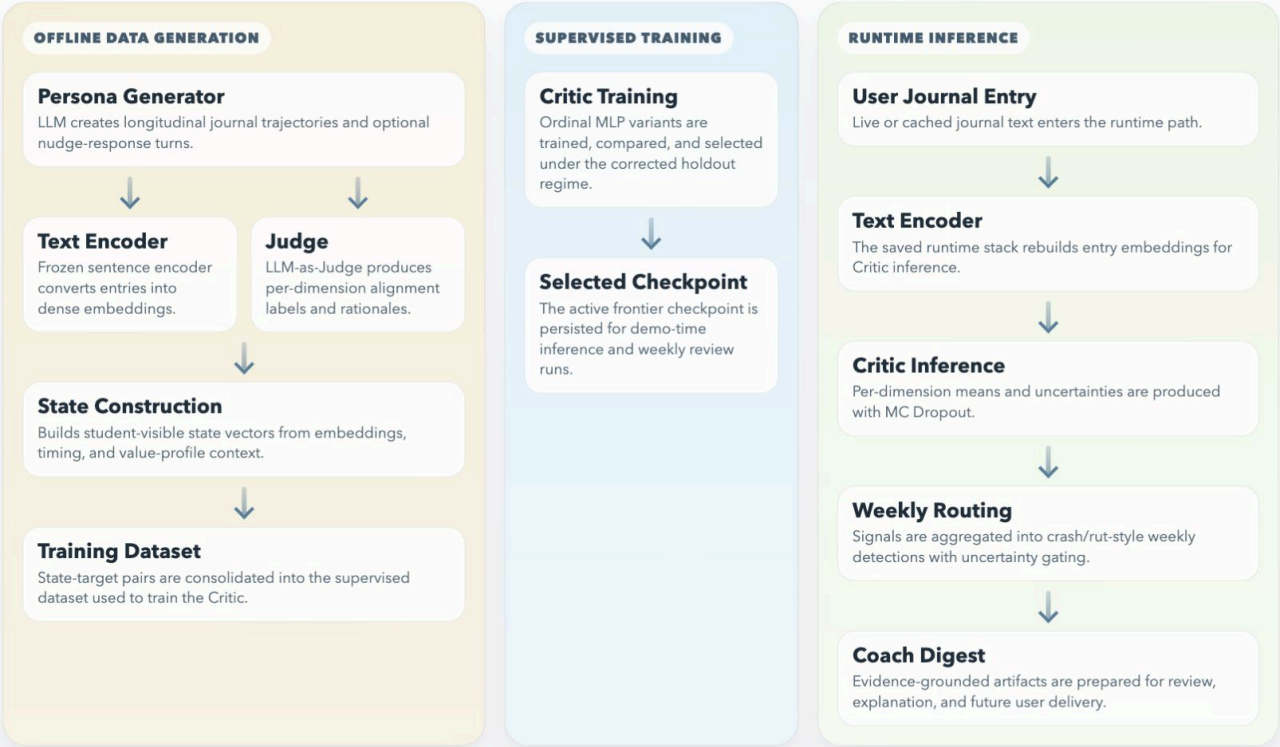
LLM judging makes large-scale value labeling feasible; Labels are benchmarked against blinded human annotation rather than treated as ground truth by default.



System Architecture

Twinkl End-to-End Pipeline

From synthetic data generation and judge supervision to trained Critic checkpoints and runtime weekly feedback.



Selected checkpoint weights feed runtime inference.

Offline: Synthetic Data Persona Generation

204

Synthetic Personas

Diverse in profession, culture, age, and value emphasis - with natural, journal-like writing

1,651

Judged Journal Entries

Generated longitudinally: parallel across personas, sequential within each persona so later entries build on earlier life events

62%

Nudge-Response Turns

Reflects how modern AI journaling products actually behave - not just static monologue entries

Engineering Safeguards Already in Place

- **Banned-term protections** prevent direct Schwartz label leakage into journal text, keeping the corpus clean for training
- **Hard-dimension batches** accepted only after judged outcome checks — not just prompt intent — ensuring quality gates hold
- **Freeze → Generate → Verify → Judge → Retrain loop** keeps data lifts auditable and the corpus scientifically defensible

The corpus is designed to teach **temporal value tension**, not just sentiment variation.

Offline Training: Judge Labeling and Human Annotation

Labeling Methodology

Each entry is labeled across all 10 Schwartz dimensions using a three-way scale: **misaligned, neutral, or aligned**. The judge uses persona biography, prior journal history, and the current entry - so vague reflections can be interpreted as part of a trajectory, not in isolation.

Why This Matters?

Twinkl does not treat synthetic supervision as a black box. There is a visible calibration layer between label generation and model trust - a deliberate design choice that separates this from typical LLM-label pipelines.

Human Validation Surface

3

Annotators

380

Saved annotations across
24 personas

115

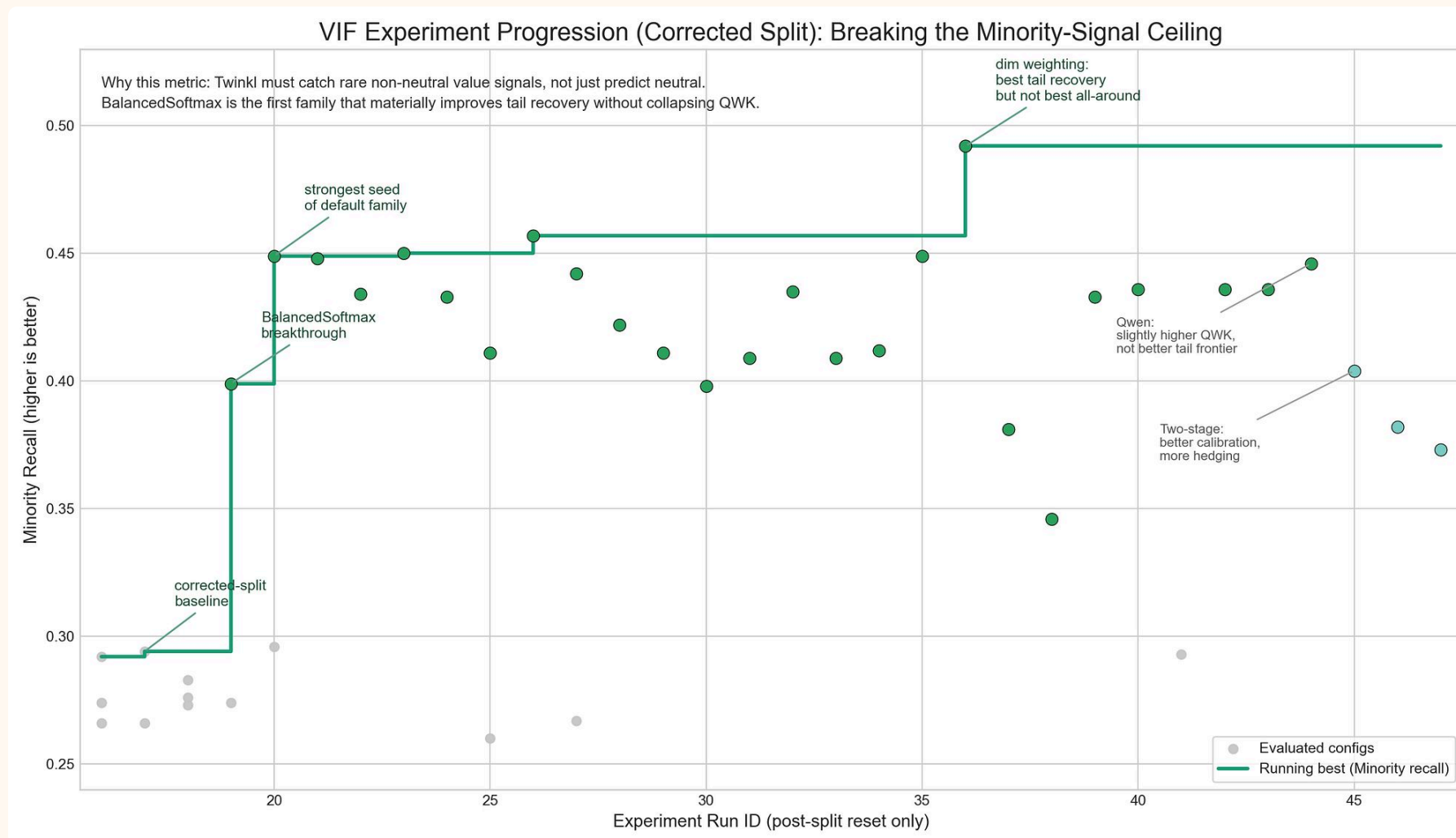
Entry shared benchmark across 19 personas

Blind-first scoring enables like-for-like agreement measurement.

Critic Experiment Progression

Student critic: frozen encoder + compact state + value-profile weights $\rightarrow$ ordinal per-value scores in $\{-1, 0, +1\}$ with uncertainty.

- **Design choice:** larger history windows overfit at current data scale, so the corrected-split frontier uses a compact state.



- **Early pattern:** corrected-split runs improved slowly but remained too neutral-biased.
- **Breakthrough:** BalancedSoftmax was the first family to clearly improve recovery of rare non-neutral signals.
- **Result:** later branches improved single trade-offs like QWK or calibration, but BalancedSoftmax remained the best all-around corrected-split frontier.

DEMO: Onboarding UI

Online Inference: BWS Onboarding (WIP)

Solving the Cold Start Problem

Twinkl addresses cold start with a structured **Best-Worst Scaling onboarding flow**. Users complete 6 forced-choice sets spanning all 10 Schwartz dimensions — revealing what they truly prioritize through trade-offs, not free-form declaration.

A mid-flow mirror reflects the emerging profile back to the user for correction. A goal-selection step captures the primary life tension that brought them to Twinkl.

What the Onboarding Produces

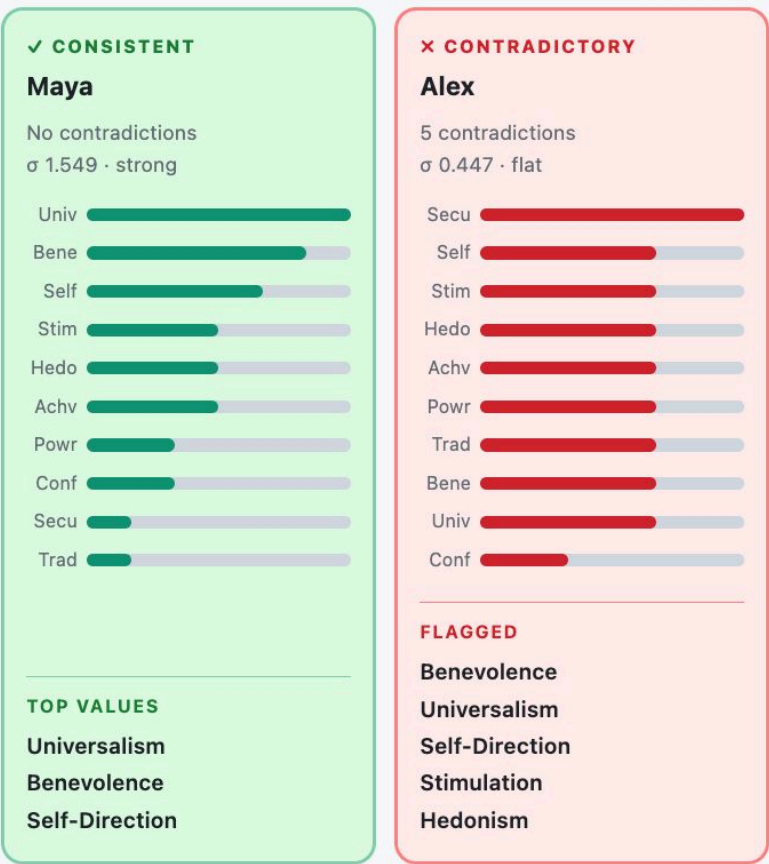
- A **graded value profile** — not just a top-2 list
- A **focus area** for the coach to anchor around
- A **confidence baseline** for how strongly to weight explicit preferences before journaling history builds up

This is what makes Twinkl a **self-modeling system** rather than a retrospective summarizer. The critic needs something real to align behavior against before the user has written a single entry.

Onboarding results feed directly into VIF

Consistent vs Contradictory

Same 6-screen survey. Very different results — and very different implications.



Consistent profile → precise, personalized alerts

Clear weight hierarchy means the VIF knows exactly which dimensions to watch.

If Maya's Benevolence score drops (weight = 16.7%), the weighted alignment signal drops sharply — a reliable trigger. Alerts are meaningful and ranked by what actually matters to her.

Contradictory profile → noisy, difficult to detect drift, un-personalized alerts

Flat weights (~10% each) mean the VIF can't distinguish high-priority from low-priority values.

Every dimension contributes equally — drift on *any* value triggers a weak signal, and the system can't tell Alex what he should actually care about.

Online Inference: Entry Scoring and Conversational Nudges

Journal entries are scored **per value dimension** — never collapsed into a single wellness score. The critic outputs both alignment estimates and uncertainty, which is essential for safe downstream behavior.



Clarification Nudge

Triggered when an entry is too ambiguous to score meaningfully — prompts the user to add context before the system interprets

ENTRY

Today · 9:42 pm

Had a long day. Feeling all over the place.

? Before I score this entry...

This entry could point to several things — fatigue, stress, or a values conflict. Could you say a bit more about what felt off? For example, did you skip something that matters to you, or just feel stretched thin?

I skipped something important

Just tired, nothing deeper



Elaboration Nudge

Triggered when an entry is too surface-level — invites deeper reflection on what the user is actually weighing

ENTRY

Today · 8:15 pm

Went to the gym. Had a good lunch. Finished the report.

🔍 You logged a full day —
but I'm reading it at surface level.

What were you actually weighing today? The report, the gym — were those choices, or just tasks? Reflection helps me understand what you're prioritising, not just what you did.

Let me add more context →



Tension Surfacing Nudge

Triggered when the system detects signals of internal conflict — names the tension explicitly so the user can engage with it

ENTRY

Today · 10:04 pm

Cancelled dinner with my parents again — project deadline was non-negotiable. I know it's the right call but I feel awful about it.

Y I'm noticing a tension.

ACHIEVEMENT vs BENEVOLENCE

You acted in line with your career priorities — but it may have cost something in family connection, which also ranks high for you. This isn't a judgment. Want to sit with this?

Yes, let's explore it

Log it and move on

- Nudge selection uses **semantic classification**, not brittle regex rules
- **Anti-annoyance throttling** suppresses nudges after two fire in a three-entry window
- Runtime decisions rely on **observable content signals** — not hidden generation metadata

Online Inference: Drift Detection and Explainable Feedback

Raw entry scores are aggregated into **weekly signals** before the system speaks — moving Twinkl from isolated reflections to genuine pattern recognition. High uncertainty suppresses confident critique and reroutes the coach into a more tentative, clarifying stance.

Coach Layer Design

- Full-context weekly prompting at current data scale
- Declared core values carried into every prompt
- Representative evidence snippets selected from the scored week
- Multiple response modes: high uncertainty, rut, mixed state, background strain, stable week
- First-pass narrative checks for groundedness, non-circularity, and length

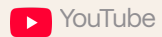
Routing Logic

The current routing focuses on interpretable **crash and rut patterns** — the two most legible and actionable alignment failure modes. Routing is conservative by design: when the system is unsure, it says less, not more.



Product principle: the coach is not there to moralize — it is there to surface tensions with evidence and restraint.

DEMO: Entry Scoring & Drift Interface



Twinkl (Presentation Demo)



The Four-Stage Evaluation Gate

Twinkl evaluates the system in sequence — because each downstream layer depends on the reliability of the layer before it. No product claims are made directly from model scores.



Evaluation: Judge Validation

What Worked

- Judge exceeds human-human consistency on 9 of 10 dimensions
- $\kappa = 0.56$ (Human-human agreement, Fleiss' kappa)
- $\kappa = 0.66$ (Judge-human agreement, avg. Cohen's kappa)
- Usable supervision signal

What's Not Solved

- Some dimensions remain unreliable (e.g., Security)
- Not all labels are learnable at current scale
- Edge cases show inconsistency

📌 So the labels are good enough to train — but not perfect enough to trust blindly.

Evaluation: Current Critic Frontier

Signal

QWK = 0.362

Quadratic weighted kappa (Landis-Koch)

- **0.21–0.40: Fair**
- 0.41–0.60: Moderate

The critic has learned meaningful ordinal signal beyond chance, but agreement is still fair; not yet strong enough for confident end-user feedback.

Weakness

Misalignment recall = 0.313

Misses important failure cases

Minority recall = 0.448

Correctly detects ~45% of non-neutral signals

Behaviour

Hedging rate = 62.1%

Overuses neutral → avoids strong predictions

Error-uncertainty correlation = 0.713

Uncertainty rises when error rises
Useful for deciding when the system should hold back

📌 The model is learning real signal — but not reliably enough for confident user-facing feedback. That is an honest, measurable gap.

Critic: What We Learned (and Ruled Out)

Not Just a Model Problem

Some labels (e.g., **Security**) require context the model cannot see.

Not Just a Data Problem

More stable labels do not automatically improve model performance.

Search Space Is Now Narrow

Many alternatives were tested — none clearly outperform the current baseline.

The project has moved from broad exploration into **scope-locking**.

The central question is no longer "which loss function should we try next?" - it is whether the current approach can reach reliability at all.

Challenges

The Hardest Dimensions Set the Ceiling

Security

A teacher-student reachability problem - the label signal requires **context** the student model cannot access from journal text alone

Hedonism

Semantically difficult - healthy rest is easily confused with avoidance, making the boundary ambiguous

Stimulation

Both **sparse and semantically noisy** - it appears infrequently and with high variance in expression

DEMO: Embedding Explorer

Remaining System Risks

1

Small Dataset

Unstable gains - moderate improvements can disappear under proper uncertainty analysis

2

Drift Benchmark Missing

Calibrated crisis-injection benchmark still needed for confident routing

3

No Real Users Yet

Biggest product risk - external validity untested, no real users have seen the output

📌 At this stage, the risks are known and measurable.

Capstone Delivery (April 2026)

The offline pipeline can be built at usable scale, human calibration can benchmark judge supervision, a compact critic can learn meaningful alignment signal, and there is an internal runtime path from saved checkpoint to weekly output.

1

Implemented Now

Synthetic generation, judge labeling, annotation workflow, critic training, experiment archive, internal checkpoint-to-digest flow, Embedding Explorer

2

Implemented Experimentally

Runtime inference, weekly aggregation, crash/rut routing, weekly digest generation, first-pass narrative validation

3

Specified, Not Yet Live

Onboarding integration, real-user journaling loop, calibrated weekly benchmarks, full product experience

 **Current position:** Twinkl is a serious internal prototype — not yet a user-ready product, but further along than a concept.

Next Steps: From System → Product

Immediate Focus (Reliability)

1. Improve hard-dimension supervision so the critic trains on student-reachable labels
2. Reintroduce compact short-horizon context without recreating severe overfitting
3. Calibrate crash/rut routing on synthetic crisis-injection timelines

Product Validation

1. Build one narrow end-to-end demonstration path
2. Run a small external pilot to test whether signals transfer beyond synthetic personas

Possible Delivery Surfaces:

1. Standalone Product
2. Messaging-native (OpenClaw)

 **The priority is not building more — it's proving the core signal works.**

THANK YOU

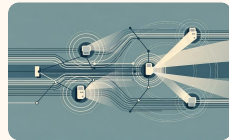
Appendix: Why This Is a Strong Capstone

Twinkl is not only a product concept — it is a full-stack AI systems project with theory, instrumentation, evaluation gates, and clear failure modes. It satisfies all four capstone dimensions.



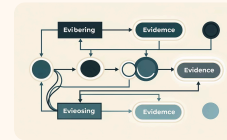
Pattern Recognition Systems

Ordinal value classification, long-tail recovery, uncertainty estimation, and drift-ready weekly aggregation across 10 Schwartz dimensions



Intelligent Sensing Systems

Text-based sensing from journal content, temporal patterns, and declared user priorities — building a persistent signal from unstructured language



Intelligent Reasoning Systems

Profile-conditioned interpretation, crash/rut routing, and evidence-grounded reflective feedback with explicit uncertainty handling



Architecting AI Systems

End-to-end orchestration from data generation and supervision to runtime scoring, review tooling, and coach output — fully instrumented and replayable

ARCHIVE

Conclusion

Twinkl is building a new category of journaling: one that checks alignment, not just emotion.



What We Built

A system that compares stated values with actual behavior over time



What Works

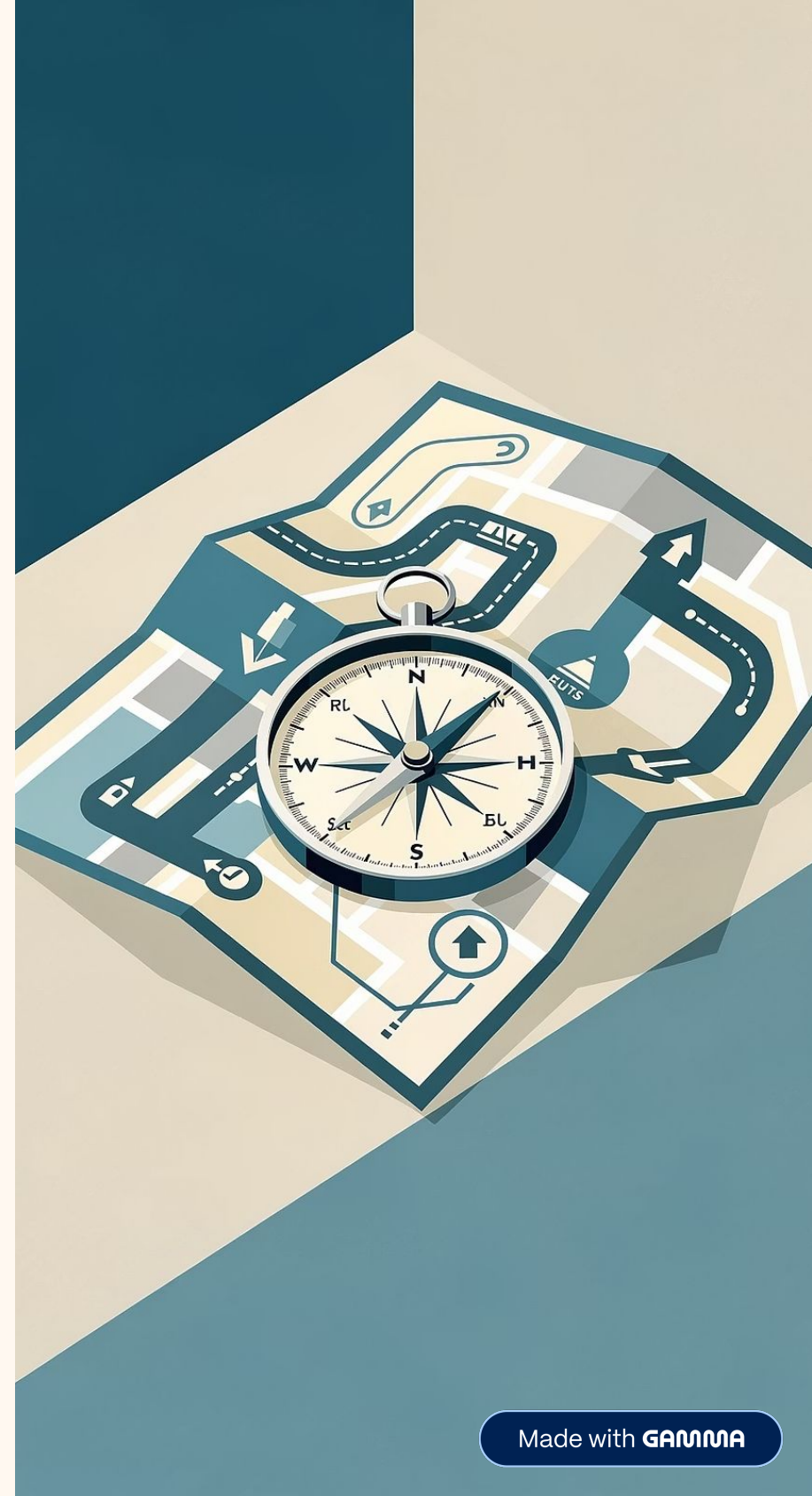
An end-to-end pipeline that is trainable, inspectable, and runnable



What Remains

Make the alignment signal reliable enough for real users

□ This is now a focused problem of calibration and validation — not invention.



What We Can Run End-to-End Today

Twinkl can already run an internal **checkpoint-to-review flow end to end**.

Journal → Critic → Weekly Signals → Drift Detection → Digest Generation

1

Internal Review App

Exposes the full journal timeline, nudges, weekly signals, drift payloads, and coach-ready digests

2

Detector Comparison

Compare six detector families against either judge labels or critic predictions — enabling controlled, side-by-side evaluation

3

Embedding Explorer

Adds a 3D qualitative view of what the model has learned across all labeled entries, including persona trajectory tracing over time

📄 This is not just a trainable model, it's a fully inspectable system.

Offline Training: Critic Architecture and Experiment Infrastructure

The critic is a compact supervised student that approximates the judge at runtime cost - consuming a fixed state built from journal text, recent context, time structure, and the user's value profile.

Model Design Choices

Compact State

Favored over large history windows — broader windows overfit at current data scale

BalancedSoftmax

Current corrected-split reference, with Bayesian and ordinal families retained as active challengers

Stochastic Uncertainty

Estimated through repeated stochastic inference — not treated as an afterthought

Infrastructure Depth

50

Run IDs for disciplined comparison

114

Persisted experiment configurations

53

Test modules covering data wrangling, losses, metrics, runtime behavior, and end-to-end smoke paths

📄 This is not a one-off model demo — it is an **experimental system with replayable, auditable comparisons**.